What genome-scale Perturb-seq responses contain: an independent-screen analysis and the limits of reading perturbation responses as regulatory networks

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Abstract

Perturbation screens are increasingly used both to benchmark expression-prediction models and to infer gene regulatory networks (GRNs). Here, we ask what a CRISPRi knockdown profile in the Replogle 2022 K562 screens actually contains, and how much of it can be interpreted as network structure.

A statistical decomposition of 4,507 genome-wide knockdown pseudobulk profiles splits each profile into four parts: a mean ("generic") direction, a few shared low-rank axes, a residual, and the sampling noise expected from non-targeting controls. Pooled over knockdowns, expected noise is 47% of profile energy; for the median knockdown it is 74% (47–50% and 74–78% across noise-constant estimators). How the remaining signal divides between shared axes and the residual depends strongly on the number of axes removed: the residual is 78% of signal with none removed, 51% with three and 18% with fifty. When fitted separately in the independent essential-gene screen, the generic direction agrees, and the two three-dimensional shared subspaces share two closely aligned directions, which load on ribosomal and mitochondrially encoded transcripts.

We score held-out predictions against an *independent-screen specific-response target*: the same knockdown measured in the other screen, with the shared structure removed. On this target, the best simple predictor (similarity transfer) reaches a correlation of 0.071 [0.062, 0.080]. That is about 11% of the ceiling implied by cross-screen agreement under an equal-reliability assumption. On a magnitude-sensitive metric (R^2 relative to predicting zero), predictors are near zero or negative on independent-screen targets; after training-only scalar calibration the best specific-target value is 0.004 [0.003, 0.005] (similarity transfer, $k = 100$).

A test of whether the knockdown's own responders relay its effect ("cascades") gives a clear signal on raw and generic-removed profiles. On the specific residual it finds nothing (difference from a random-mediator null -0.004 $[-0.012, 0.004]$). In semi-synthetic positive controls the pipeline detects an injected cascade component carrying 5% of the signal energy when the mediators are among the knockdown's strongest measured responders, but it loses most or all sensitivity when mediators are ranked arbitrarily or do not respond visibly at the transcript level. The absence of a residual cascade signal is therefore informative only about cascades through strongly responding measured transcripts.

Promoter binding (1 kb upstream, 197 TFs) shows no detectable discrimination of knockdown responders under our responder definition. This holds for measured ChIP-seq binding as well as for motif occupancy, so the failure is not attributable to the occupancy approximation.

Across contexts (Norman 2019; Wessels 2023), strong responses shared by both contexts rarely reverse sign (1.4% and 1.1%). Gene-level context differences (nominal $|z_{\text{diff}}| > 3$) occur at 8.5× and 3.0× the rate expected under a noise-matched null; reversals are 1.8% and 1.1% of these, and the remainder are changes in size, gains, losses and intermediate-strength changes.

Together, these results bound what perturbation responses can say about direct regulatory edges. Simulation studies of model components (memory-based hidden-regulator detection, intervention-induced indirect edges, prior forms, propagation depth) are reported in the Supplement as estimator-specific results.

1 Introduction

Large Perturb-seq screens [1] are used for two purposes that are easy to conflate. One is *response prediction*: predicting the transcriptional profile of an unseen perturbation. The other is *network inference*: reading direct regulatory edges out of perturbation responses. Benchmarks have found that sophisticated network-inference methods barely beat simple baselines [2]. Deep response predictors often do not beat linear models or the average perturbation profile [3]. Part of this has been traced to systematic variation shared across perturbations, which inflates apparent performance [4].

Here we characterise what the profiles contain before asking either question. The Replogle K562 data include a genome-wide screen and a separate essential-gene screen, which share 1,454 strongly knocked-down targets. Because the screens are independent experiments, sampling noise in one is independent of the other. Comparing them measures reproducible structure (agreement between screens) without a noise model; interpreting the disagreement as noise still requires assumptions. The screens are not exact replicates, however: they were sequenced at different times after transduction (day 8 vs day 6 in the original description [1]), so disagreement between them also includes biological differences.

Contributions, and how they relate to prior work. Systema [4] established that shared perturbation variation inflates apparent predictive performance and proposed evaluating perturbation-specific effects. Our additions are: (1) a decomposition of genome-scale knockdown responses into expected noise, a generic direction, shared low-rank axes and a residual, with sensitivity analyses and a cross-screen check of both the noise model and the axes; (2) an *independent-screen* specific-response target with an explicit reliability derivation, used to compare simple predictors on correlation and on a magnitude-sensitive metric; (3) a cascade ("domino") test of whether knockdown-specific responses are relayed through the knockdown's measured responders, with a matched positive control; (4) a promoter-scoped test separating failures of motif occupancy from failures of measured binding to predict knockdown responses; (5) a noise-matched analysis of context-dependent responses (gained, lost, shared and reversed) in two combinatorial datasets. Simulation studies of GRN model components, which began this project, are moved to the Supplement and stated as results about the estimators and regimes tested.

2 Results

2.1 A statistical decomposition of knockdown responses

We decomposed 4,507 genome-wide knockdown profiles (pseudobulk z-scores; knockdowns with fold expression < 0.5 and energy-test $p < 0.05$; target gene masked). Each profile x is split as

$$x = (x \cdot u)\,u \;+\; V^{\top}V(x - (x \cdot u)u) \;+\; r,$$

where u is the unit mean knockdown profile (the generic direction), V holds the top K principal axes of the generic-removed matrix, and r is the residual. Expected sampling noise for a profile from n cells is cG/n, with c estimated from 585 non-targeting control profiles. We call the components *generic*, *shared axes* and *specific*. These are statistical labels, not biological ones: shared axes are themselves outputs of regulatory networks, and the residual may contain technical structure.

How much is noise. Pooled over knockdowns, expected noise is 47% of total profile energy. The median knockdown is 74% noise (Table 1, Fig. 1a). The pooled figure is lower because a minority of strong knockdowns carries most of the energy. Restricting to the 2,000 most variable genes lowers pooled noise to 25%. These fractions are stable to the choice of noise-constant estimator (pooled 0.47–0.50, median 0.74–0.78 across profile-level median, mean and trimmed mean and gene-specific estimators) and to resampling of the 585 controls (95% intervals 0.467–0.473 and 0.733–0.742; Supplementary Table S11). These intervals reflect only uncertainty in the noise constant, not misspecification of the noise model. Allocating expected noise to components in proportion to their dimensionality ($(G-1-K)/G$ to the residual) assumes isotropic noise. Because the shared axes are fitted to the same noisy matrix, they absorb somewhat more than their share of noise, so the specific signal is, if anything, slightly underestimated.

How the signal divides depends on K. The generic share of signal is stable (0.22). The residual's share falls from 0.78 ($K = 0$) to 0.51 ($K = 3$, the permutation-null choice) and 0.18 ($K = 50$) (Fig. 1b). Shares were broadly similar across tertiles of cell count and knockdown strength; the residual share of signal ranged 0.48–0.61 (Table 1). Knockdowns with the most cells have the lowest *absolute* expected noise (cG/n), but the highest noise *fraction* (median 0.77), because they tend to have weaker responses: knockdowns that impair growth yield fewer cells and stronger effects.

Checks of the noise model and the axes. We could not split perturbed cells into halves, because only pseudobulk profiles are available. Instead we used the two screens. For a knockdown present in both, $\mathbb{E}\|x_{\mathrm{gw}} - x_{\mathrm{es}}\|^2$ equals the sum of both screens' noise energies plus any true between-screen difference. The observed difference was $1.30\times$ the predicted

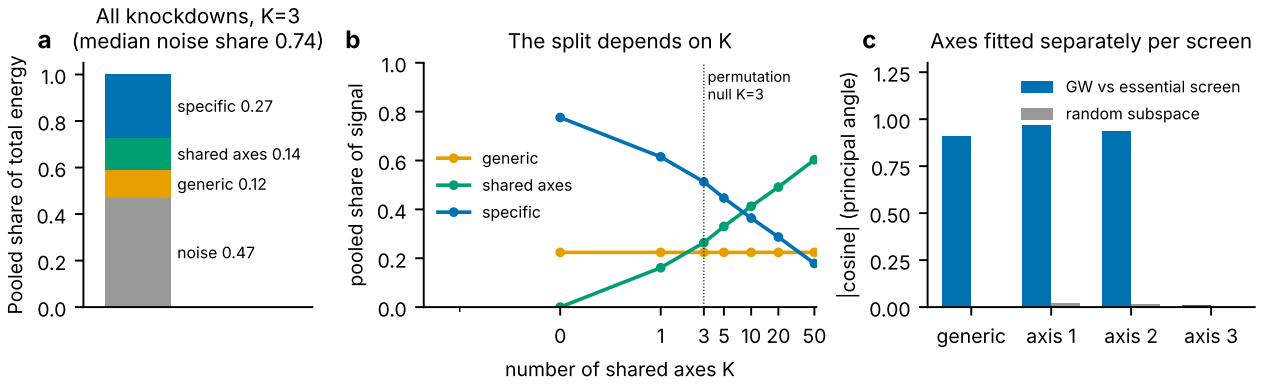


Figure 1: Decomposition of 4,507 Replogle genome-wide knockdown profiles. (a) Pooled shares of total energy ($K = 3$); the median knockdown's noise share is shown in the title. (b) Pooled shares of signal as a function of the number of shared axes removed. (c) Agreement between generic directions and shared axes fitted separately in the genome-wide and essential-gene screens (1,457 knockdowns in both), compared with a random subspace.

noise (median ratio; IQR 1.13–1.61). For random pairs of non-targeting profiles the ratio was 1.18× (Table 2). We treat these as consistency checks, not an attribution: they are medians of ratios over different sets of pairs, and the excess can reflect screen or batch effects, extra noise in perturbed profiles, or true between-screen differences. They indicate that the control-calibrated noise model is of the right order but does not account for all cross-screen disagreement. When the generic direction and three shared axes were fitted separately in each screen, the generic directions agreed ($|\cos| = 0.91$). The two three-dimensional subspaces share two closely aligned directions (principal-angle cosines 0.97, 0.94) and not a third (0.01; random subspace 0.02). Matching individual axes gives the same picture: genome-wide axis 1 matches essential axis 1 ($|\cos| = 0.95$) and axis 2 matches axis 2 (0.91), while axis 3 matches nothing (at most 0.12; Supplementary Table S12). The aligned directions load on ribosomal-protein genes and mitochondrially encoded transcripts: aligned direction 1: EEF1A1, TPT1, RPS18, RPS27, SNHG29, RPS3, RPL13A, RPS8; aligned direction 2: MT-CO3, PRG2, MT-ND4, MT-CO2, MT-CO1, MT-ATP6, GAS5, MT-CYB. The non-matching third axis loads on erythroid and haemoglobin genes. We did not run a formal enrichment analysis, and the mitochondrial direction may be partly technical.

The residual reproduces. For the 1,454 knockdowns in both screens, the specific residuals correlate with median $r = 0.22$ genome-wide (mismatched knockdowns: -0.003). On the 1,000 most variable genes used for benchmarking (below) the median is 0.48. Within the genome-wide screen, only 19 targets have a second guide or TSS; their residual agreement was low (median 0.05). Within-screen repeatability therefore cannot be separated from cross-screen agreement in these data.

2.2 An independent-screen specific-response target

Reliability. Suppose the two screens measure the same underlying response s with independent noise: $x_1 = s + e_1$ and $x_2 = s + e_2$, with reliabilities $R_i = \mathrm{Var}(s)/\mathrm{Var}(x_i)$. Then

$$\mathrm{corr}(x_1, x_2) = \sqrt{R_1 R_2}, \qquad \max_f \mathrm{corr}(f, x_2) = \mathrm{corr}(s, x_2) = \sqrt{R_2}.$$

Cross-screen correlation is therefore not itself a ceiling. Under the additional assumption $R_1 = R_2$, the best achievable correlation with screen 2 is $\sqrt{r_{12}}$. If the screens' true responses differ (for example because of timing), r_{12} underestimates the reliability of each screen, and $\sqrt{r_{12}}$ is a conservative (low) estimate of the ceiling for that screen's own response. Independence removes *shared sampling noise* from the comparison, so a model cannot be rewarded for fitting the same noise realisation it was trained on. It does not remove noise from the target. It also cannot remove dependencies shared by both screens: the same cell line, very probably the same guide designs (guide identities are not in the released pseudobulk files, so we could not verify this), and the same processing pipeline.

Protocol. Five folds grouped by target gene. Within each fold:

Table 1: Decomposition sensitivity. Pooled quantities are sums over knockdowns divided by total (or signal) energy; "median noise share" is the per-knockdown median. "Specific" = residual energy minus its expected noise (can be biased by the noise-model error in Table 2).

Setting	Pooled share of total energy				Pooled share of signal			Median noise share / KD
	noise	generic	shared axes	specific	generic	shared axes	specific	
Number of shared axes K (all genes, 4,507 knockdowns)								
$K = 0$	0.47	0.12	0.00	0.41	0.22	0.00	0.78	0.74
$K = 1$	0.47	0.12	0.09	0.33	0.22	0.16	0.62	0.74
$K = 3$ (permutation null)	0.47	0.12	0.14	0.27	0.22	0.26	0.51	0.74
$K = 5$	0.47	0.12	0.18	0.24	0.22	0.33	0.45	0.74
$K = 10$	0.47	0.12	0.22	0.19	0.22	0.41	0.36	0.74
$K = 20$	0.47	0.12	0.26	0.15	0.22	0.49	0.29	0.74
$K = 50$	0.47	0.12	0.32	0.09	0.22	0.60	0.18	0.74
Gene panel ($K = 3$)								
all genes	0.47	0.12	0.14	0.27	0.22	0.26	0.51	0.74
2000 most variable	0.24	0.18	0.23	0.35	0.24	0.30	0.46	0.53
mean UMI > 1 (1728 genes)	0.30	0.14	0.26	0.30	0.21	0.37	0.43	0.57
Cells per knockdown, tertiles ($K = 3$; 1 = fewest)								
tertile 1	0.46	0.13	0.15	0.26	0.24	0.28	0.48	0.67
tertile 2	0.48	0.11	0.13	0.29	0.21	0.24	0.55	0.77
tertile 3	0.53	0.08	0.11	0.29	0.17	0.23	0.60	0.77
Knockdown strength, tertiles ($K = 3$; 1 = strongest)								
tertile 1	0.55	0.09	0.12	0.25	0.19	0.26	0.55	0.80
tertile 2	0.48	0.10	0.15	0.27	0.20	0.28	0.52	0.72
tertile 3	0.41	0.15	0.15	0.29	0.26	0.26	0.49	0.65

Table 2: Noise-model and axis checks across the two screens.

Check	Value
Observed / predicted noise energy, gw−es difference, same knockdown (1394)	1.30 [1.13, 1.61]
Same, random pairs of non-targeting profiles (300)	1.18 [0.94, 1.55]
$\lvert\cos\rvert$ between generic directions fitted separately per screen	0.91
Principal-angle cosines, 3 shared axes fitted separately per screen	0.97, 0.94, 0.01
Same for a random 3-dimensional subspace	0.02, 0.02, 0.00

- The generic direction and shared axes are refitted on training knockdowns only (K by permutation null; 2–4 across folds).
- The gene panel is the 1,000 most variable genes among training knockdowns, chosen per fold.
- Targets are the held-out knockdown's genome-wide profile, its essential-screen profile (raw, mean-centred by the training mean, and specific residual), with the target gene masked.

Similarity transfer predicts a held-out knockdown of gene t as a weighted average of training knockdown profiles. Weights are the correlation between gene t's own expression column and each training target's column, computed across training knockdowns only. This feature uses only training data and is available at prediction time. The number of neighbours $k \in \{10, 20, 50, 100\}$ was not tuned; all values are reported. We report Pearson correlation with bootstrap 95% CIs over held-out knockdowns. As a magnitude-sensitive metric we report $R_0^2 = 1 - \|y - \hat{y}\|^2 / \|y\|^2$ (relative to predicting zero). For constant predictors correlation is undefined and shown as n/a.

Results. On raw targets the mean training profile scored 0.340 (same screen) and 0.247 (independent screen), close to the best predictor (similarity transfer, $k = 100$: 0.374 and 0.285; Table 3, Fig. 2). Mean-centring the independent-screen target

Table 3: Held-out knockdown prediction (1,396 held-out knockdowns, 5 folds). Correlation with bootstrap 95% CI; R_0^2 = R-squared relative to predicting zero (magnitude-sensitive). n/a: correlation undefined for a constant prediction. Bottom rows: cross-screen agreement of the targets on the same panel and the implied ceiling under equal reliability.

Predictor	Same screen, raw corr [95% CI]	R_0^2	Indep., raw corr [95% CI]	R_0^2	Indep., mean-centred corr [95% CI]	R_0^2	Indep., specific corr [95% CI]	R_0^2
zero	n/a	-0.000	n/a	-0.000	n/a	0.000	n/a	-0.000
mean training profile	0.340 [0.329, 0.350]	0.042	0.247 [0.237, 0.257]	0.007	n/a	0.000	-0.003 [-0.005, -0.000]	-0.000
kNN k=10	0.310 [0.297, 0.322]	-0.189	0.244 [0.232, 0.256]	-0.189	0.108 [0.094, 0.122]	-0.174	0.048 [0.040, 0.056]	-0.120
kNN k=20	0.342 [0.330, 0.354]	-0.093	0.265 [0.253, 0.277]	-0.117	0.116 [0.102, 0.130]	-0.106	0.054 [0.046, 0.063]	-0.056
kNN k=50	0.369 [0.356, 0.380]	-0.025	0.283 [0.271, 0.295]	-0.064	0.130 [0.115, 0.144]	-0.056	0.064 [0.056, 0.073]	-0.017
kNN k=100	0.374 [0.362, 0.386]	0.003	0.285 [0.273, 0.297]	-0.039	0.128 [0.114, 0.143]	-0.033	0.071 [0.062, 0.080]	-0.004
factorised (generic mean + pathway kNN50 + specific kNN50)	0.356 [0.343, 0.369]	0.014	0.277 [0.264, 0.289]	-0.017	0.135 [0.122, 0.148]	-0.016	0.064 [0.056, 0.073]	-0.017
Cross-screen r of the target (mean)	–	–	0.59		0.58		0.47	
Ceiling $\sqrt{r}$ (mean; equal reliability)	–	–	0.74		0.74		0.66	

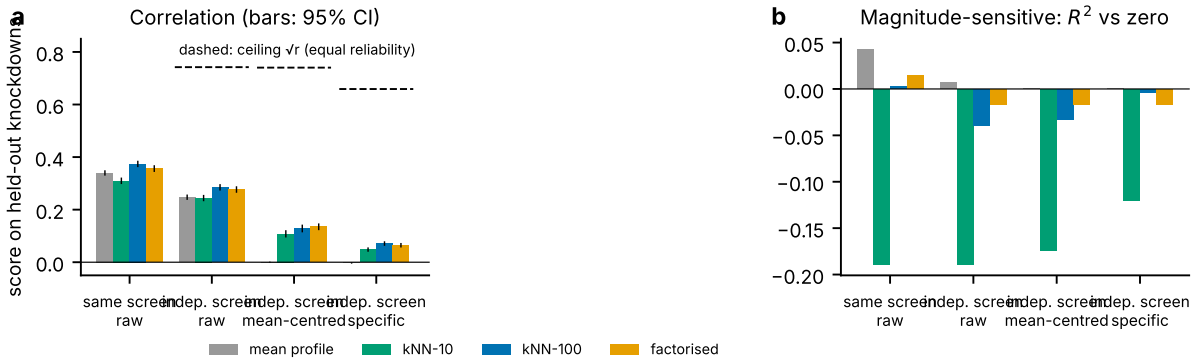


Figure 2: Independent-screen evaluation. (a) Correlation of four predictors with four targets; dashed lines mark the equal-reliability ceiling $\sqrt{r}$ on the same gene panel. (b) R^2 relative to predicting zero.

removes the mean profile's credit by construction. Among substantive predictors it gives 0.108–0.135, with the factorised predictor best (0.135 [0.122, 0.148]). On the specific residual, similarity transfer with $k = 100$ was best (0.071 [0.062, 0.080]). Under equal reliability that is about 11% of the ceiling $\sqrt{r}$ (0.66 on this panel). Uncalibrated, only the mean profile achieved positive R_0^2 on an independent-screen target (raw profiles, 0.007); small-k predictors were strongly negative (-0.12 to -0.19). Positive correlation with negative R_0^2 can reflect poor amplitude calibration rather than an absence of signal, and averaging over more neighbours shrinks predictions. We therefore calibrated each predictor with a scalar $\hat{y}_{\text{cal}} = \lambda\hat{y}$, with $\lambda \in [0, 10]$ chosen on training knockdowns only (leave-one-out predictions for 300 training knockdowns per fold, maximising mean R_0^2; $\lambda = 0$ allowed). After calibration (Supplementary Table S14), all predictors had positive R_0^2 on independent raw profiles (0.033 for the mean profile to 0.047 for $k = 100$), small positive values on mean-centred targets (0.005–0.013, factorised best) and on specific targets (0.002–0.004; $k = 100$: 0.004 [0.003, 0.005]). Fitted factors were below 1 ($\bar{\lambda} = 0.10$–0.46 for similarity predictors), so all predictions were over-scaled. Paired per-knockdown differences show that more neighbours helps beyond shrinkage on the specific target: $k = 100$ minus $k = 10$ gives Δcorr $+0.023$ [0.016, 0.030] and, after calibration, ΔR_0^2 $+0.0024$ [0.0016, 0.0031] (Supplementary Table S15). $k = 100$ and $k = 50$ differ little (Δcorr $+0.007$ [0.002, 0.011]; calibrated ΔR_0^2 $+0.0006$ [0.0002, 0.0010]). The specific target therefore discriminates among substantive predictors, but the explained energy is very small.

2.3 Cascade test: is the specific response relayed through the knockdown's own responders?

If knockdown j acts largely *through* its strongest responders k, its profile should be predicted by $\hat{x}_j = \sum_k (a_{jk}/d_k)\, x_k$. Here a_{jk} is j's effect on k, d_k is k's own knockdown depth, and x_k is k's knockdown profile, which already contains everything downstream of k. We scored $\hat{x}_j$ against j's profile in the independent screen, excluding j and its mediators from scoring. The null keeps the same weights but uses random mediators. Mediators are restricted to the 521 genes with a measurable self-knockdown, and we used the 507 targets whose specific residual reproduces across screens (cross-screen $r > 0.3$). The estimator sums total effects, so overlapping downstream paths can be counted more than once; it is a linear, one-step reconstruction.

Table 4: Cascade test on real data (507 reproducible knockdowns, 10 mediators; bootstrap 95% CIs over knockdowns).

Profiles used (mediators and target)	corr, cascade [95% CI]	corr, random mediators	difference	frac. beats null
specific residual (original)	-0.010 [-0.015, -0.003]	-0.006 [-0.012, 0.000]	-0.004 [-0.012, 0.004]	0.47
generic removed only	0.119 [0.103, 0.136]	-0.014 [-0.024, -0.003]	0.133 [0.112, 0.153]	0.71
raw profiles	0.084 [0.068, 0.100]	-0.054 [-0.068, -0.040]	0.137 [0.119, 0.157]	0.75

Raw vs residual targets. The choice of profile determines the result (Table 4, Fig. 3a). On raw profiles, and on profiles with only the generic direction removed, the cascade prediction clearly beats random mediators: difference $+0.137$ [0.119, 0.157] and $+0.133$ [0.112, 0.153]. On the specific residual it does not (-0.004 [-0.012, 0.004]). Cascade-like structure is therefore present, but it lies along the shared axes that residualisation removes. A knockdown's strongest responders have knockdown profiles that resemble its own along those axes. This is consistent with relay through mediators, but equally with co-membership of a complex or pathway; the test cannot distinguish them.

Positive control. We built semi-synthetic data matched to the real data and ran them through the identical pipeline.

- For target j with true mediator set $\mathscr{M}_j$ (10 mediators) and weights w_k, let $c_j = \sum_{k\in\mathscr{M}_j} w_k r_k$ (real mediator residual shapes r_k) and let o_j be j's own real residual shape, orthogonalised against c_j. The noise-free target is $s_j = \|r_j\|\,(\sqrt{\alpha}\,\hat{c}_j + \sqrt{1-\alpha}\,\hat{o}_j)$, with hats denoting unit vectors. Because $\hat{c}_j \perp \hat{o}_j$, α is exactly the fraction of the target's signal energy along the cascade direction; the realised fraction is reported for each run (Supplementary Table S13).
- Independent Gaussian noise is added for each screen and for each mediator profile, at each profile's real cross-screen reliability.
- "Coverage" hides a fraction of the true mediators from the candidate pool, mimicking mediators that were never knocked down.

In the base generator the true mediators are the target's 10 strongest measured responders, the weights are the real a_{jk}/d_k, and the target's values at the mediator genes are set to its real effects on them (so the mediators are visible as responders). Under this generator the pipeline detected $\alpha = 0.05$ at all coverage levels (Table 5, Fig. 3b), and the real-data upper bound (0.004) lies below the $\alpha = 0.05$ lower bound even at 25% coverage (0.005). Sensitivity depends strongly on the generator (Supplementary Table S13, Fig. 3c):

- heterogeneous mediator strengths and halving either screen's reliability reduced but did not remove sensitivity;
- drawing the true mediators at random among measured genes (ranking errors) reduced sensitivity sharply (at coverage 1: 0.006 at $\alpha = 0.05$, 0.023 at $\alpha = 0.5$);
- when the mediators' coordinates in the target were not set to visible responses, the test had no detectable sensitivity even at $\alpha = 0.5$;
- a combination of adverse settings also removed all sensitivity.

Bounded conclusion: the pipeline detects a nominal 5% injected cascade component under the tested semi-synthetic generator, including reduced mediator coverage, provided the mediators are among the knockdown's strongest measured transcript responders. It is not a calibrated exclusion test across plausible generators. In particular, cascades transmitted through mediators whose transcripts respond weakly, or through changes not visible at the transcript level (for example protein activity), are not excluded. The reliability-normalised share statistic $\max(\text{corr},0)^2/r$ is strongly attenuated by noise (0.11 at $\alpha = 1$), so we do not use it as an effect size.

2.4 Promoter binding does not predict knockdown responders

Scope: 1-kb upstream promoters of 7,839 Perturb-seq-measured genes in K562; ENCODE ChIP-seq (197 TFs); responders defined as genes with $|z| \geq 0.5$ in the TF's strongest knockdown (22 TFs with fold expression < 0.85 and ≥ 20 responders). Distal regulation, TF activity changes and timing are outside this setup.

Motif occupancy weighted by ATAC accessibility predicted ChIP binding better than the maximum motif score (mean per-TF AUROC 0.610 [0.599, 0.621] vs 0.566 [0.555, 0.576]; paired difference 0.045 [0.041, 0.049], higher for 97% of

Table 5: Positive control, base generator: difference between cascade and random-mediator correlations (mean [95% CI]) for semi-synthetic targets with exact cascade share α (fraction of signal energy) and mediator coverage, run through the same pipeline as Table 4. Robustness to other generators: Supplementary Table S13.

True cascade share α	coverage 1.0	coverage 0.5	coverage 0.25
0	0.000 [-0.003, 0.003]	-0.000 [-0.003, 0.003]	-0.002 [-0.005, 0.001]
0.05	0.047 [0.044, 0.049]	0.026 [0.023, 0.029]	0.008 [0.005, 0.011]
0.1	0.066 [0.063, 0.069]	0.038 [0.035, 0.041]	0.013 [0.010, 0.016]
0.25	0.106 [0.102, 0.110]	0.061 [0.057, 0.065]	0.026 [0.022, 0.030]
0.5	0.153 [0.147, 0.158]	0.087 [0.083, 0.092]	0.039 [0.035, 0.043]
Real data (specific residual)		-0.004 [-0.012, 0.004]	

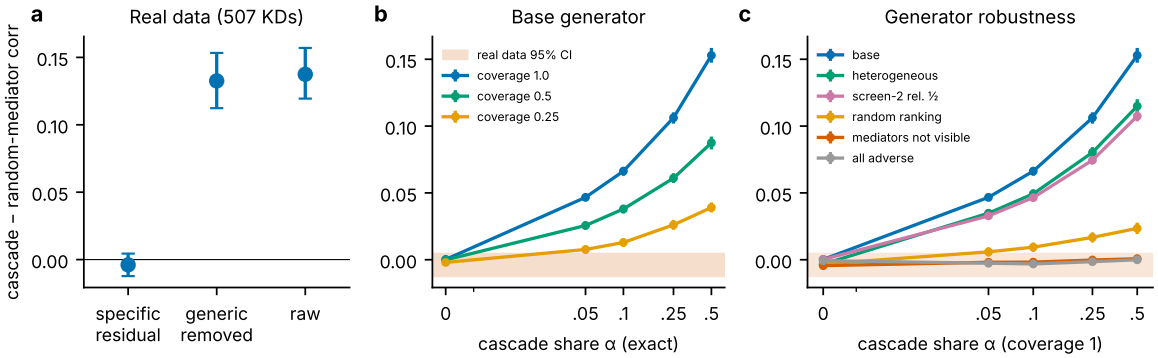


Figure 3: Cascade test. (a) Real data: cascade prediction minus random-mediator null for three choices of profile. (b) Positive control, base generator: the same statistic for semi-synthetic data with exact cascade share and varying mediator coverage; shaded band = real-data 95% CI on the specific residual. (c) Sensitivity under alternative generators (coverage 1).

TFs) and better than accessibility alone (+0.033 [0.022, 0.044]) (Table 6, Fig. 4). A leave-TF-out "hub" score (the fraction of other TFs binding the promoter) predicted binding far better (0.761 [0.753, 0.768]). The hub score correlates moderately with ATAC signal (Spearman 0.33) and expression (0.23), and keeps its AUROC within accessibility or expression quintiles (0.753 and 0.758). Its performance is therefore not fully explained by these coarse stratifications; broadly co-occupied promoters or assay-related structure (for example highly ChIP-able regions) are plausible contributors, but we did not test them directly.

No score predicted knockdown responders: occupancy 0.518 [0.497, 0.542], occupancy×ATAC 0.492 [0.468, 0.520]. Crucially, the same was true of *measured* ChIP binding (0.496 [0.473, 0.511]), including within expression quintiles (0.488). The failure is therefore not attributable to the occupancy approximation. Measured promoter binding shows no detectable discrimination of responders under this responder definition ($|z| \geq 0.5$ in the strongest knockdown); other definitions, distal elements or larger TF sets could behave differently.

2.5 Context: shared strong responses keep their sign, but many responses change

We compared the effect of perturbing A alone ($\bar{x}_A - \bar{x}_{\text{ctrl}}$) with its effect in C's context ($\bar{x}_{AC} - \bar{x}_C$). This was done for 262 ordered pairs in Norman 2019 [5] (K562, CRISPRa; 2,000 variable genes) and 106 in Wessels 2023 [6] (THP-1, Cas13 knockdown). These are comparisons of transcriptional *responses*, which combine direct regulation, indirect paths, feedback and state changes. Preserved response signs do not imply preserved signs of direct edges.

Comparing "significant in one context, not in the other" is dominated by power: under a null in which both contexts share the same true effect, many genes switch between significant and non-significant by chance. In Wessels, for example, the null predicts more "lost" calls than observed (Table 7). We therefore tested each gene directly for a context difference, $z_{\text{diff}} = (d_2 - d_1)/\sqrt{s_1^2 + s_2^2}$. The null is matched to each comparison's sampling noise: both contexts share the precision-weighted mean effect, and each estimate is redrawn with its own standard error (20 replicates). $|z_{\text{diff}}| > 3$ is a nominal per-gene threshold without multiple-testing control. We use it to compare aggregate counts with the matched null, which is strong evidence for context dependence overall, not to certify individual genes.

Table 6: Promoter binding and knockdown response (per-TF AUROC, bootstrap 95% CI over TFs).

Score	mean per-TF AUROC [95% CI]	TFs
Predicting ChIP-seq binding at promoters		
PWM_max	0.566 [0.555, 0.576]	197
occupancy (mu=Ethr-2)	0.568 [0.557, 0.579]	197
occupancy x ATAC	0.610 [0.599, 0.621]	197
ATAC only	0.577 [0.569, 0.585]	197
hub (leave-TF-out binding frequency)	0.761 [0.753, 0.768]	197
Within-stratum AUROC (averaged over 5 quintiles)		
hub (leave-TF-out binding frequency) | within ATAC quintile	0.753 [0.745, 0.761]	197
occupancy x ATAC | within ATAC quintile	0.591 [0.581, 0.602]	197
PWM_max | within ATAC quintile	0.566 [0.556, 0.576]	197
hub (leave-TF-out binding frequency) | within expression quintile	0.758 [0.750, 0.766]	197
occupancy x ATAC | within expression quintile	0.610 [0.599, 0.622]	197
PWM_max | within expression quintile	0.565 [0.555, 0.576]	197
Predicting knockdown responders ($\|z\| \geq 0.5$)		
ChIP bound (measured binding)	0.496 [0.473, 0.511]	22
occupancy (mu=Ethr-2)	0.518 [0.497, 0.542]	22
occupancy x ATAC	0.492 [0.468, 0.520]	22
PWM_max	0.512 [0.493, 0.535]	22
ChIP bound | within expression quintiles	0.488 [0.470, 0.502]	22
occupancy x ATAC | within expression quintiles	0.494 [0.469, 0.522]	22

Each significant change is assigned to one category (z_1, z_2: effect z-scores alone and in context): *reversal*, $|z_1| > 3$ and $|z_2| > 3$ with opposite signs; *same sign, other size*, both > 3 with the same sign; *gained*, $|z_1| < 1.96$ and $|z_2| > 3$; *lost*, $|z_1| > 3$ and $|z_2| < 1.96$; *other (intermediate strength)*, everything else (at least one effect between 1.96 and 3 in absolute value).

Significant context differences occurred at 8.5× the null rate in Norman (12,084 vs 1,421) and 3.0× in Wessels (1,692 vs 562) (Table 7, Fig. 5). As a fraction of all significant changes (Norman; Wessels): reversals 1.8%; 1.1%; same sign, other size 23.3%; 4.6%; gained 26.0%; 28.7%; lost 25.3%; 29.4%; other (intermediate strength) 23.6%; 36.3%. As a fraction of strong responses shared by both contexts ($|z| > 3$ in both), reversals were 1.4% (218 of 15,778) and 1.1% (18 of 1,657). In Shifrut 2018 [7] (T cells, with vs without stimulation), knockout effects were too weak to test: split-half replicates shared only 6–12 strong targets across 18 knockouts (Supplementary Table S10). **Conclusion:** among strong responses present in both contexts, signs are preserved for ≈99%. However, context-dependent changes in aggregate, including gains and losses of responses, are common and well above noise, so context does not merely rescale a fixed set of responses.

3 Discussion

Three conclusions follow, each with stated limits.

First, most of a typical knockdown profile is sampling noise, and much of the signal lies on a generic direction and a few shared axes. Two of these shared directions reproduce across screens and load on ribosomal and mitochondrially encoded transcripts. How much signal is "specific" is not a fixed number: it depends on how many shared axes one removes. Benchmarks and network-inference methods should report their choice and its sensitivity.

Second, scoring against the independent screen's specific response is a stricter target than raw profiles. It discriminates among substantive predictors, and simple predictors recover only a small fraction of its estimated ceiling. Unlike Systema [4], which re-scores predictions within a dataset, the independent screen protects against rewarding a model for fitting the same noise realisation. The ceiling derivation makes explicit what cross-screen agreement does and does not bound.

Third, perturbation responses are poor direct readouts of regulatory edges in these data. Measured promoter binding does not predict responders. The knockdown-specific residual shows no detectable relay through the knockdown's strongly responding measured transcripts, a test with demonstrated sensitivity only under a favourable generator. Relay-like structure is visible along shared axes, where cascades and shared module membership cannot be told apart. Response-level context dependence is common even though reversals of strong responses are rare. Models that map perturbation responses to edges therefore need either additional data that resolve direct regulation (for example acute perturbations, time courses, or

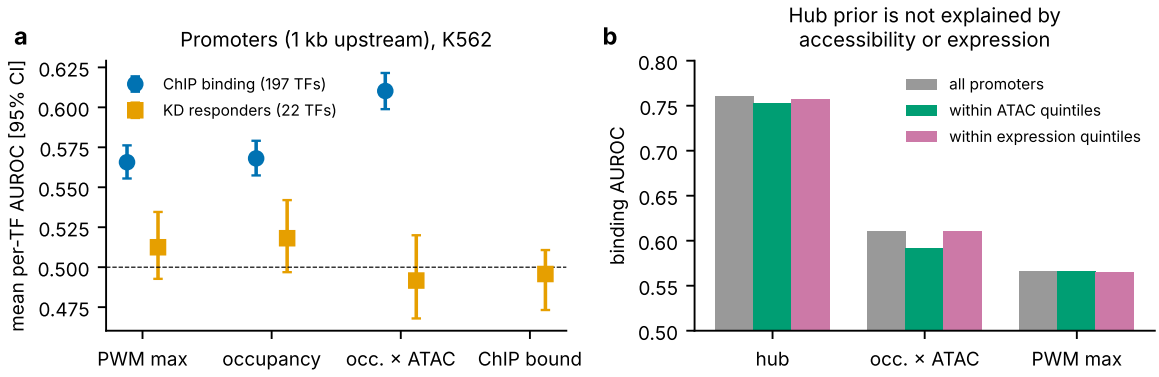


Figure 4: Binding at promoters. (a) Binding and response prediction by motif, occupancy and measured binding. (b) The hub prior keeps its accuracy within accessibility and expression strata.

Table 7: Context-dependent responses vs a noise-matched null (counts of gene × comparison, summed over comparisons; null mean ± s.d. over 20 replicates). The null shares a precision-weighted mean effect across contexts, which makes it generous to "gained"/"lost" counts (conservative for claiming context dependence).

	Norman 2019 (262 comparisons)		Wessels 2023 (106 comparisons)	
Gene-level category (summed over comparisons)	observed	matched null	observed	matched null
strong alone ($\|z\| > 3$)	64,962	77,544 ± 186	7,848	10,432 ± 79
strong in context	26,311	28,060 ± 79	5,168	7,963 ± 75
shared, same sign	15,560	20,647 ± 75	1,639	2,807 ± 41
shared, reversed sign	218	3 ± 2	18	1 ± 1
gained (alone $\|z\| < 1.96$)	7,256	4,033 ± 50	2,714	3,462 ± 53
lost (context $\|z\| < 1.96$)	36,891	39,618 ± 169	4,844	5,215 ± 51
significant change, $\|z_{\text{diff}}\| > 3$	12,084	1,421 ± 34	1,692	562 ± 18
of which reversal	218	3 ± 2	18	1 ± 1
of which same sign, other size	2,813	50 ± 8	77	3 ± 1
of which gained	3,136	553 ± 15	486	161 ± 10
of which lost	3,061	227 ± 18	497	129 ± 12
of which other (intermediate strength)	2,856	589	614	269

direct binding-and-response assays), or explicit components for shared response structure.

Limitations.

- All Replogle analyses use pseudobulk z-scores. Split-half validation of the noise model on perturbed cells was not possible, and the cross-screen check indicates the noise model is approximate.
- The two screens differ in timing, and probably share guide designs and processing, so cross-screen agreement mixes noise with biology and may share technical dependencies.
- The cascade test is linear and one-step. Its sensitivity is demonstrated only under generators in which mediators are visible as strong transcript responders; it is not a calibrated exclusion test.
- Binding conclusions are limited to proximal promoters in K562.
- Context analyses compare responses, not edges, and the null uses estimated effects as truth.
- The simulations in the Supplement concern our own simulator and estimators; the regulator-mask benefit disappears when a quarter of true regulators are missing from the annotation (Supplement S7).

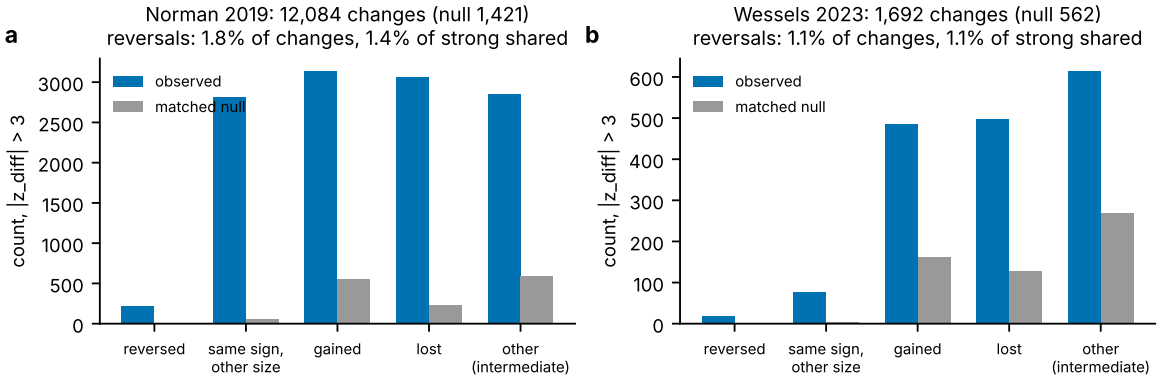


Figure 5: Composition of significant context-dependent changes (nominal $|z_{\text{diff}}| > 3$) in (a) Norman 2019 and (b) Wessels 2023, against a noise-matched null. All five categories are shown; they sum to the total.

4 Methods

Replogle data. K562 genome-wide and essential-gene normalised pseudobulk files (`K562_gwps_normalized_bulk_01`, `K562_essential_normalized_bulk_01`; `.h5ad`), Replogle et al. 2022 (figshare release accompanying the paper). The gene-level values are the released normalised (z-scored relative to controls) pseudobulk expression changes. Knockdowns: not non-targeting, fold expression < 0.5, energy-test $p < 0.05$ (4,507). The target gene's own column is set to 0. Essential-screen profiles were mapped to shared genes (8,200) and the strongest knockdown per target used.

Decomposition. The generic direction $u = \bar{x}/\|\bar{x}\|$. Shared axes are the top-K right singular vectors of the generic-removed matrix (randomized SVD). The permutation null is the top singular value after shuffling each gene's column independently. Noise energy is cG/n with $c = \text{median}(\|x\|^2 n/G)$ over non-targeting profiles; the residual's expected noise is scaled by $(G-1-K)/G$. Pooled shares are sums over knockdowns; medians are per knockdown. Cross-screen axis stability uses principal-angle cosines (singular values of $V_a V_b^\top$).

Benchmark. As described in Section 2.2. Bootstrap CIs use 2,000 resamples of held-out knockdowns.

Cascade test. Mediators: knocked-down genes with raw self z < -1 (521). Responders: top-K mediators by |screen-1 profile at the mediator's column|. Weights: a_{jk}/d_k, with a_{jk} the raw z of j's knockdown on gene k and d_k the self z of k's knockdown. Targets with cross-screen residual $r > 0.3$ (507). Scoring excludes j and the selected mediators. Null: random mediators with the same weights. Positive control: as in Section 2.3; noise s.d. per profile set so that its expected reliability equals the real cross-screen r (clipped to [0.05, 0.95]).

Binding. Promoters: 1 kb upstream of refGene TSSs (hg38). Energies: JASPAR 2024 [8] PWM log-odds, scanned on both strands. E_{thr} is the 10^{-4} quantile of scores on background sequences with promoter base composition; occupancy $= \sum_{\text{sites}} \sigma(\mu - E)$ at $\mu = E_{\text{thr}} - 2$, and the accessibility-weighted version adds the log ATAC signal at each position. ChIP: ENCODE K562 conservative IDR peaks [9] (accessions in `data/raw/encode_manifest.tsv`). Hub: fraction of the other 196 TFs whose peaks overlap the promoter. Expression: Replogle gene-level mean UMI.

Context. Norman 2019: log-normalised (CP10k) single-cell matrix, 2,000 most dispersed genes, perturbed genes masked. Wessels 2023: scPerturb h5ad [10], same normalisation, perturbations with ≥ 60 cells (singles) and ≥ 100 (pairs). Standard errors are $\sqrt{\text{Var}/n}$ for each group mean.

Uncertainty. Bootstrap over knockdowns, TFs or targets as stated. Paired comparisons between predictors use per-knockdown differences. Seeds for simulations are listed in the Supplement.

Code and data availability. Code, results, environment specifications and the commands used are at https://github.com/kylerzook/cross-screen-perturbseq (release tag `v1.2`; `README.md`, `run_all.sh`). The main-text pipeline (decomposition, noise and axis checks, binding from cached promoter features, Norman preprocessing and context analysis, benchmark and cascade test) was re-run from a fresh clone in a clean environment built from `requirements.txt`; all re-computed values matched the reported results exactly, except Monte Carlo null replicates in the context analysis, whose comparison order depended on Python's hash seed; the order is now fixed and the output verified to be deterministic (`docs/13_clean_run_check.md`). Commands for the Supplement simulations are supplied as used but were not re-verified end to end. An archived release with a DOI will be added on acceptance. Data identifiers are listed in `data/README.md`: the Replogle figshare files; ENCODE accessions; JASPAR 2024 CORE vertebrates non-redundant; MotifCentral v1.0.0 (ProBound); scPerturb h5ad files for Wessels and Shifrut; Norman 2019 (GEO GSE133344).

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Supplementary material

The analyses below started the project. They concern specific estimators, simulators and regimes, and their conclusions are stated at that level. All simulation results use the seeds listed. "Diagnostic re-implementations" of published method families (PHOENIX-, DCDI-, CellOracle- and GENIE3-style) implement only each method's core idea in our code. They are controls, not evaluations of the original software.

Metric definitions (all simulations). The candidate-edge universe is every ordered pair $(j \to i)$ of distinct *measured* genes: $70 \times 69 = 4{,}830$ pairs in the 80-gene simulator, of which about 114 are true edges. AUROC and average precision (AUPRC) are computed over this universe. AUPRC× is AUPRC divided by the base rate (true edges / candidates), so 1 = random. E is the number of true edges among measured genes; "top-E" metrics use the E highest-scored pairs. "Indirect" pairs have a directed path of length ≥ 2 (through any gene, measured or hidden) and no direct edge.

S1. Thermodynamic occupancy model

Occupancy calibration. For each TF, E_{thr} is the 10^{-4} quantile of PWM energies on 400,000 background positions with promoter base composition (a FIMO-like threshold). Occupancy uses $\mu = E_{\text{thr}} + \{-4, -2, 0\}$ (-2 pre-registered as default). ProBound models (MotifCentral v1.0.0; human model preferred, lowest fit id) were rescaled to a maximum log-affinity of 0.

T1, synthetic genome (1,500 genes, 20 JASPAR TFs, 3 seeds). Two different comparisons are reported.

- Over all targets, summed occupancy at $\mu = E_{\text{thr}}$ reached AUROC 0.959 vs 0.820 for the maximum motif score.
- Restricted to weak-cluster targets vs non-targets, occupancy reached 0.968 vs 0.647.

A fitted occupancy model recovered TF activities μ (Spearman 0.976).

T3, cell-state μ (K562 vs HepG2). Test R^2 for log expression: 0.172 (accessibility only), 0.185 (fixed μ), 0.188 (state-specific μ). For the HepG2−K562 *difference*, R^2 rose from 0.006 (fixed μ) to 0.017 (state-specific μ), an improvement of 0.011. The inferred $\Delta\mu$ did not correlate with independent proxies (Spearman 0.06–0.09, permutation $p > 0.29$).

ProBound vs JASPAR (122 TFs). Binding AUROC was higher with ProBound for 69–73% of TFs (paired Wilcoxon $p < 10^{-7}$), by about 0.02. There was no gain for function (13 TFs).

S2. Latent neural ODE on a three-gene cascade, and a 10-gene toy

Simulation. A→B→C with Hill kinetics ($K = 0.6$, $n = 2$), Euler–Maruyama ($dt = 0.005$). 200 cells (150 train), 41 observations at interval 0.2, measurement noise 5% or 20% of the per-gene s.d. **Model.** Gene-aligned latent states (residual per-gene MLP encoder and decoder) and dynamics $dz/dt = W\sigma(a(z-c)) - \text{softplus}(g)z + b$ with $\text{diag}(W) = 0$. Free latent units are initialised from a learned constant ("from start") or from $(x_t, x_t - x_{t-1})$. Loss: K-step rollout MSE (4 Euler substeps per interval) + reconstruction + $\lambda\|W\|_1$. Adam, learning rate 10^{-2}, cosine schedule, 2,000 iterations (1,200 for from-start runs). λ grid $\{0, 10^{-5}, \ldots, 10^{-2}\}$; 3 seeds. An edge is called if its strength ($|W_{ij}| \times$ range of j's activation along the rollout) is $\geq 10\%$ of the largest. **Results.** With all genes observed, the exact graph was recovered in 57–71% of settings. With B hidden, the shortcut A→C was called in 81–100% of fits. With knockdowns of the hidden B supplied and modelled (applied to a latent unit with a learned efficiency), A→ h →C was recovered without the shortcut in 75% of settings (latent–B correlation 0.975). With the same knockdowns unmodelled, the shortcut was always called. In a 10-gene toy, a binding prior raised AUROC from 0.749 to 0.947 (0.962 with knockdown-aware fitting). Two indirect edges stayed in the top 9 until the intermediate TF was itself knocked down.

S3. 80-gene simulator: ablations and experiment selection

Simulator (in-house, SERGIO-style): 80 genes (16 TFs, 8 non-TF regulators, 56 targets; about 140 edges; TF cascades of depth ≥ 3; 70% activators). Per-edge Hill coefficient $n \in \{1..4\}$ and threshold $K \sim U(0.5, 2)$; decay $U(0.5, 1)$; process noise 0.05; a hidden cell-size factor; Poisson counts with log-normal library size; noisy labelling velocity (1× s.d.); 10 unmeasured genes; 1,500 wild-type cells and 80 cells per knockdown, with efficiency $U(0.05, 0.3)$. Seeds 0–9. **BOND-V:** nodewise adaptive lasso [11] (BIC) of velocity on Hill features $\phi(x_j) = x_j^2/(K_j^2 + x_j^2)$, $K_j = \text{median}(x_j)$, plus a decay term. Features are standardised before division by the penalty factor $1 - \alpha p_{ij}$. Each gene's own-knockdown cells are dropped from its equation. Ablation results (AUPRC×, 10 seeds):

- full model 18.6; without prior 15.5; ignoring interventions 17.6; static (no velocity) 16.4;
- knockdown mean-difference 26.4, but with 36% indirect edges in its top E;
- diagnostic re-implementations: CellOracle-style 8.9, GENIE3-style [12] 5.7.

Disambiguation-driven knockdown selection beat random and hub-based choice in 10/10 seeds (area under the AUPRC× coverage curve $+0.95 \pm 0.67$ vs random). It was not tested on real data.

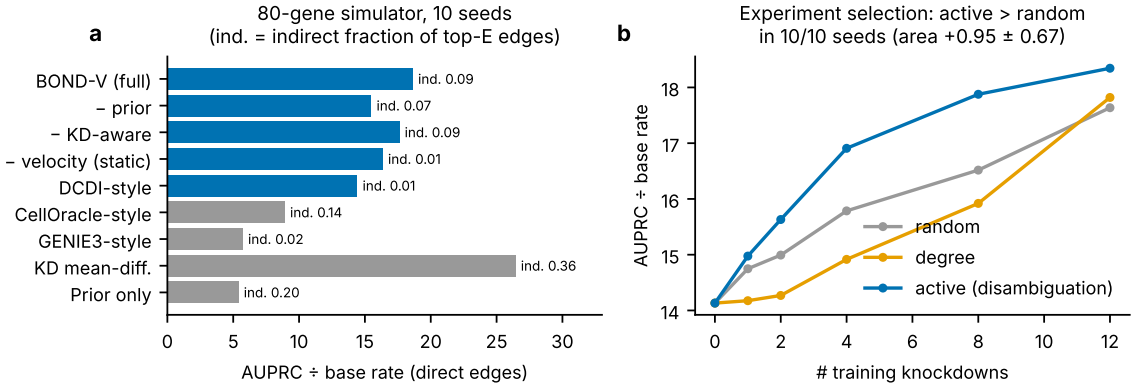


Figure S1: 80-gene simulator. (a) Ablations and diagnostic re-implementations. (b) Knockdown selection strategies.

S4. Linear SEM on real held-out knockdowns

On 203 strongly knocked-down Replogle genes, a knockdown-aware steady-state linear SEM predicted held-out knockdown profiles with correlation 0.012 ± 0.298, against 0.422 ± 0.402 for the mean training profile. ChIP and motif priors did not change this.

S5. A lag-feature ridge detector of hidden regulators (memory)

Scope. This tests one estimator: ridge regression of an observed gene's 5-step-ahead change on the current observed state, with and without lags of 1, 2, 4 and 8 samples. It is not a test of memory-function methods in general [13–15]. **Simulation.** Measurement-noise levels are relative to each gene's simulated within-trajectory fluctuation s.d. They are not calibrated against any real assay, so we do not call any level "realistic". **Results.** With regulator process noise four times that of other genes and clean data, the detector ranked children of hidden regulators above other genes (AUROC 0.84 vs 0.51 with nothing hidden; Table S1). It failed with weaker regulator fluctuations, with measurement noise $\geq 0.5\times$ the fluctuation s.d., and with sampling intervals ≥ 0.25. Estimates of the hidden regulator's timescale and of the number of hidden regulators (lagged-residual rank) failed.

S6. Intervention-induced indirect edges (errors-in-variables)

For A→B→C with $B = bA + \varepsilon_B$, $C = cB + \varepsilon_C$ and B measured as $\tilde{B} = B + e$, regressing C on $(A, \tilde{B})$ gives $\beta_{A\to C} = cb\lambda$, $\lambda = \sigma_e^2/(\sigma_e^2 + \mathrm{Var}(B|A))$ [16]. Perturbing A leaves β unchanged but makes it easier to detect. Perturbing B lowers λ (Table S2). In the 80-gene simulator, perturbing the source raised the indirect-call rate only slightly, and hidden mediators dominated (Table S3). A simple identifiability flag was modestly informative (Table S4).

S7. Prior forms and imperfect regulator masks

With an *exact* regulator list, a hard identity mask raised AUPRC× from 15.5 (no prior) to 22.2, and adding a soft site-level binding prior on top gave 23.2 (Table S5). The reading is "mask first; a site prior adds a little on top". A hard binding-site mask capped recall at 25% and was harmful. With imperfect masks (Table S6, Fig. S3), each missing regulator cost far more than each false one, because a missing regulator's true edges cannot be recovered at all.

Table S1: Lag-feature ridge detector (AUROC; 80-gene time courses; seeds 0–9 or 0–5).

Detector / condition	AUROC (mean ± sd over seeds)	seeds
Memory gain, 2 hidden regulators (clean)	0.836 ± 0.169	10
Same genes, nothing hidden (control)	0.512 ± 0.114	10
Snapshot detector ($1 - R^2$), hidden regulators	0.374 ± 0.068	10
Memory gain, weak regulator fluctuations	0.543 ± 0.127	10
Memory gain, measurement noise 0.1× fluct. sd	0.875 ± 0.129	6
Memory gain, measurement noise 0.25× fluct. sd	0.705 ± 0.151	6
Memory gain, measurement noise 0.5× fluct. sd	0.487 ± 0.052	6
Memory gain, measurement noise 1.0× fluct. sd	0.444 ± 0.058	6
control (nothing hidden), noise 0.25	0.545 ± 0.103	6
control (nothing hidden), noise 0.5	0.535 ± 0.050	6
Noise 0.5 + causal smoothing (attempted fix)	0.531 ± 0.105	6
Sampling interval 0.05 (clean)	0.807 ± 0.072	4
Sampling interval 0.1 (clean)	0.881 ± 0.072	4
Sampling interval 0.25 (clean)	0.664 ± 0.138	4
Sampling interval 0.5 (clean)	0.463 ± 0.253	4
Sampling interval 1.0 (clean)	0.436 ± 0.024	4
Sampling interval 2.0 (clean)	0.433 ± 0.037	4

Median memory gain (fraction of held-out error removed by the past), clean main condition: genes with a hidden regulator 0.027 ± 0.023, other genes -0.007 ± 0.002. Oracle gain from adding the true hidden state: 0.093 ± 0.054 vs -0.001 ± 0.000.

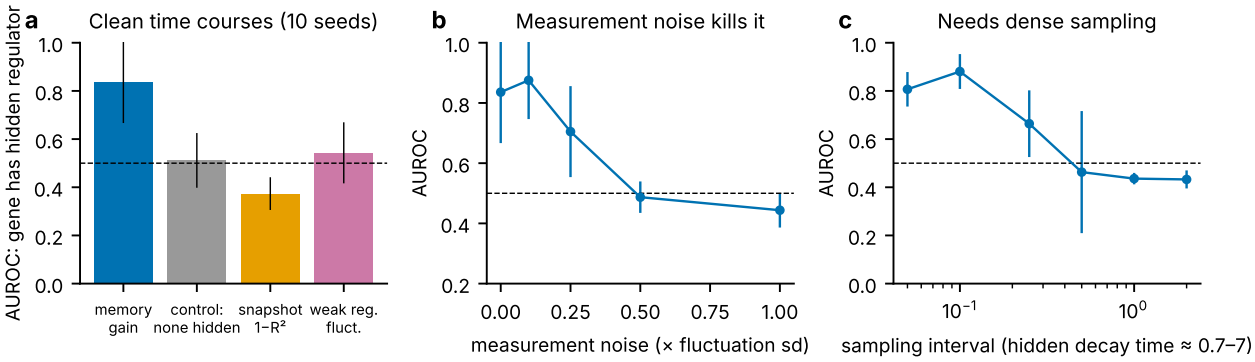


Figure S2: Lag-feature ridge detector. (a) Clean, dense time courses. (b) Measurement noise relative to simulated fluctuations. (c) Sampling interval.

- Dropping 25% of regulators (6 of 22) brought AUPRC× down to the no-prior level (15.9 vs 15.5); dropping 50% fell well below it (11.2).
- Adding false regulators degraded performance gradually: 19.1 with 50% extra, and 15.9 only with 200% extra (44 false regulators).
- A mask with both errors (25% missing, 100% false) was worse than no prior (10.7). Adding the soft site prior recovered only part of the loss (12.4).

A hard mask is therefore worthwhile only when the regulator annotation has high recall; with lower recall, soft priors or no prior are safer.

S8. Propagation depth: true vs estimated operator

We evaluated every regulator (seeds 0–9), stratified by its true downstream depth, and scored both operators against the *same* targets on the *same* log-normalised scale (Table S7, Fig. S4).

- **Targets:** (i) the noise-free steady-state knockdown effect, mapped to the log scale with the first-order transform of the simulator's normalisation; (ii) the measured mean effect in the simulated data.

Table S2: Three-gene check ($b = c = 1$, $\sigma_B = \sigma_C = \sigma_e = 0.5$, $n = 2{,}000$, 300 replicates).

Condition	theory $cb\lambda$	simulated coefficient	mean t	Lasso selects A→C
observational	0.500	0.497 ± 0.024	21.0	1.00
perturb A	0.500	0.500 ± 0.022	23.9	1.00
perturb B	0.100	0.099 ± 0.017	5.5	1.00
perturb A and B	0.100	0.076 ± 0.011	6.8	1.00

Table S3: Indirect pairs called as edges by perturbation status (BOND-V without prior; seeds 0–9 $\times$ 3 random halves of the perturbation set).

Source (regulator)	2-hop mediator	indirect pairs	% called (top-E), $\pm$ seed sd
source perturbed	mediator unperturbed	874	5.9 ± 3.2
source perturbed	mediator perturbed	881	4.9 ± 3.4
source perturbed	mediator hidden	175	14.9 ± 14.3
source perturbed	longer path only	1709	0.2 ± 0.3
source unperturbed	mediator unperturbed	779	5.1 ± 1.9
source unperturbed	mediator perturbed	862	4.3 ± 3.3
source unperturbed	mediator hidden	179	17.3 ± 12.5
source unperturbed	longer path only	1723	0.3 ± 0.5

- **True operator:** the linearised production Jacobian at the wild-type steady state, with its predictions mapped to log scale by the same transform.
- **Estimated operator:** BOND-V, 5-fold over knockdowns.

Remaining mismatches between the two operators include linearisation error, and cell-to-cell heterogeneity (input levels, cell size) that the true operator, computed at mean inputs, ignores. The noise-free and measured targets correlate 0.79 on average.

For regulators with downstream depth ≥ 3, the mean over $D \in \{2,3,4\}$ minus $D = 1$ (bootstrap 95% CI over 82 regulators) was:

- true operator: $+0.19$ [0.16, 0.23] against the noise-free target and $+0.07$ [0.05, 0.09] against the measured target;
- estimated operator, stabilised (rescaled to spectral radius 0.8): $+0.04$ [0.01, 0.07] against the noise-free target and -0.02 [-0.05, -0.00] against the measured target;
- estimated operator, unstabilised: strongly negative (-0.35 and -0.51).

Conclusion: deeper propagation helps with the true operator in this simulator, against both noise-free and measured targets. The estimated pipeline does not reproduce that benefit: there is no consistent improvement beyond one step after stabilisation, and collapse without it. At $D = 1$ on the measured target, the estimated operator matched or exceeded the true operator (0.70 vs 0.64 for deep regulators), consistent with the heterogeneity mismatch above. Table S8 reports the earlier comparison, in which the true operator was scored on noise-free raw-scale effects. In Table S9, the $(I-J)^{-1}$ column is an algebraic inverse. It equals the infinite propagation sum only when $\rho(J) < 1$, which holds only for the stabilised rows.

S9. Context: earlier sign-flip summary and Shifrut 2018

Table S10 gives a sign-flip summary that conditions on targets strong in both contexts; Table 7 gives the unconditional, noise-matched analysis.

S10. Additional checks for the main-text analyses

S11. Reproducibility

Environments: main-text and Supplement S5–S10 analyses were run with Python 3.13 and the versions in `requirements.txt`; Supplement S1–S4 (including the PyTorch latent-ODE runs) were run mainly on a second machine with Python 3.10.12 and

Table S4: Identifiability flag (edge $j \to i$ flagged if some unperturbed m has predicted $j \to m$ and $m \to i$ edges among the top $2E$).

Quantity	Value
n predicted edges	3426
fraction flagged	0.437
precision flagged	0.273
precision unflagged	0.441
indirect frac flagged	0.089
indirect frac unflagged	0.054
share of indirect FPs that are flagged	0.561
share of TPs flagged	0.324
precision source perturbed	0.835
precision source unperturbed	0.208
precision regulator source perturbed	0.835
precision regulator source unperturbed	0.717
n edges from unperturbed regulators	739
n edges from perturbed	873
fraction of edges from non regulators	0.529
precision flagged among regulator sources	0.711
precision unflagged among regulator sources	0.820

Table S5: Prior forms (exact regulator list; seeds 0–9; mean $\pm$ s.d.).

Prior form	recall ceiling	AUROC	AUPRC$\times$	Precision@E	indirect frac.	held-out KD corr
no prior	1.00	0.834 ± 0.021	15.483 ± 2.246	0.382 ± 0.033	0.069 ± 0.026	0.332 ± 0.139
soft site prior (a=0.8)	1.00	0.872 ± 0.018	18.638 ± 2.361	0.467 ± 0.021	0.093 ± 0.024	0.418 ± 0.163
soft identity prior (non-regulators x3 penalty)	1.00	0.906 ± 0.013	20.631 ± 2.677	0.493 ± 0.023	0.106 ± 0.036	0.416 ± 0.136
hard identity mask (TFs + other regulators)	1.00	0.916 ± 0.018	22.248 ± 2.303	0.496 ± 0.031	0.140 ± 0.052	0.394 ± 0.106
hard TF-only mask	0.80	0.835 ± 0.023	19.334 ± 2.947	0.457 ± 0.054	0.219 ± 0.068	0.232 ± 0.076
hard binding mask (TF x top-15% occupancy)	0.25	0.617 ± 0.027	7.296 ± 1.169	0.224 ± 0.016	0.188 ± 0.052	-0.001 ± 0.102
hard identity mask + soft site prior	1.00	0.924 ± 0.012	23.186 ± 2.258	0.515 ± 0.032	0.155 ± 0.058	0.397 ± 0.102

the versions in `requirements-s1-s4.txt` (torch 2.14.1+cpu); some 80-gene coverage runs used the main environment. `run_all.sh` lists the commands used, in order, starting from the raw files listed in `data/README.md`. End-to-end regeneration from a clean environment was verified only for the analyses stated in the main-text Methods; the other commands are supplied but not re-verified. Random seeds are fixed in each script (`numpy.random.default_rng` with the seeds stated above). Approximate run times on a 2-core machine: decomposition and benchmark $<$10 min; cascade test with positive control about 10 min; 80-gene simulations 1–3 h.

Table S6: Imperfect regulator masks (BOND-V; seeds 0–9). Missing regulators are dropped from the mask; false regulators are random non-regulator genes added to it.

Regulator mask	allowed regulators	recall ceiling	AUPRC×	Precision@E	held-out KD corr
exact regulator set	22	1.00	22.2 ± 2.3	0.496	0.39
missing 10% of regulators	20	0.90	19.8 ± 2.2	0.454	0.32
missing 25% of regulators	16	0.73	15.9 ± 1.5	0.405	0.22
missing 50% of regulators	11	0.50	11.2 ± 2.8	0.316	0.10
+50% false regulators	33	1.00	19.1 ± 2.1	0.433	0.40
+100% false regulators	44	1.00	17.1 ± 3.2	0.429	0.41
+200% false regulators	66	1.00	15.9 ± 2.1	0.395	0.37
missing 25% + 100% false	38	0.72	10.7 ± 1.4	0.301	0.31
missing 25% + 100% false, + soft site prior	38	0.72	12.4 ± 1.4	0.336	0.29
Reference: no prior	–	1.00	15.5 ± 2.2	0.382	0.33
Reference: soft site prior only	–	1.00	18.6 ± 2.4	0.467	0.42

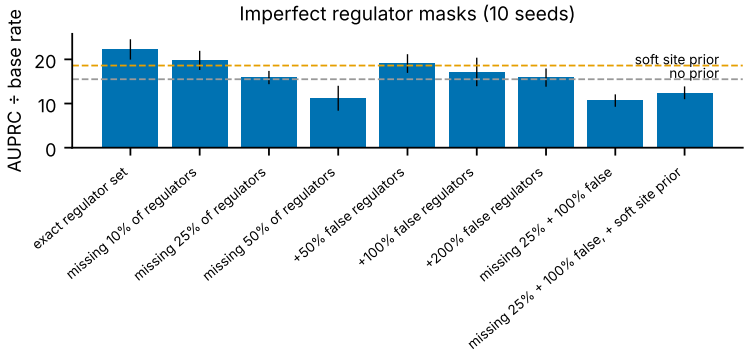


Figure S3: Imperfect regulator masks.

Table S7: Propagation depth, both operators on the same targets and scale (correlation with the knockdown effect; mean over regulators; seeds 0–9).

Operator	target	regulators (n)	D=1	2	3	4	6	8
true operator	noise-free	depth 1 (95)	0.91	0.91	0.91	0.91	0.91	0.91
true operator	noise-free	depth 2 (43)	0.78	0.88	0.88	0.88	0.88	0.88
true operator	noise-free	depth $\geq$ 3 (82)	0.57	0.73	0.77	0.78	0.78	0.78
true operator	measured	depth 1 (95)	0.71	0.71	0.71	0.71	0.71	0.71
true operator	measured	depth 2 (43)	0.71	0.75	0.75	0.75	0.75	0.75
true operator	measured	depth $\geq$ 3 (82)	0.64	0.70	0.71	0.71	0.71	0.71
estimated	noise-free	depth 1 (95)	0.39	0.18	0.06	0.04	0.01	-0.00
estimated	noise-free	depth 2 (43)	0.55	0.38	0.08	-0.00	0.05	-0.03
estimated	noise-free	depth $\geq$ 3 (82)	0.57	0.54	0.20	-0.06	0.06	-0.04
estimated	measured	depth 1 (95)	0.36	0.16	0.05	0.02	0.00	-0.01
estimated	measured	depth 2 (43)	0.58	0.37	0.07	-0.01	0.06	-0.06
estimated	measured	depth $\geq$ 3 (82)	0.70	0.54	0.12	-0.10	0.10	-0.05
estimated, stabilised	noise-free	depth 1 (95)	0.39	0.33	0.31	0.33	0.33	0.33
estimated, stabilised	noise-free	depth 2 (43)	0.55	0.53	0.50	0.50	0.51	0.51
estimated, stabilised	noise-free	depth $\geq$ 3 (82)	0.57	0.63	0.62	0.61	0.61	0.61
estimated, stabilised	measured	depth 1 (95)	0.36	0.30	0.29	0.30	0.30	0.30
estimated, stabilised	measured	depth 2 (43)	0.58	0.55	0.51	0.51	0.52	0.52
estimated, stabilised	measured	depth $\geq$ 3 (82)	0.70	0.71	0.67	0.66	0.68	0.68

Table S8: Propagation depth, earlier comparison on different targets (true operator: noise-free raw-scale effects; estimated: measured log-scale effects). Superseded by Table S7.

Operator	held-out regulators (n)	D=1	2	3	4	6	8
true operator	downstream depth 1 (95)	0.90	0.90	0.90	0.90	0.90	0.90
true operator	depth 2 (43)	0.76	0.85	0.85	0.85	0.85	0.85
true operator	depth $\geq$ 3 (82)	0.61	0.74	0.78	0.78	0.78	0.78
estimated	downstream depth 1 (95)	0.36	0.16	0.05	0.02	0.00	-0.01
estimated	depth 2 (43)	0.58	0.37	0.07	-0.01	0.06	-0.06
estimated	depth $\geq$ 3 (82)	0.70	0.54	0.12	-0.10	0.10	-0.05
estimated, stabilised	downstream depth 1 (95)	0.36	0.30	0.29	0.30	0.30	0.30
estimated, stabilised	depth 2 (43)	0.58	0.55	0.51	0.51	0.52	0.52
estimated, stabilised	depth $\geq$ 3 (82)	0.70	0.71	0.67	0.66	0.68	0.68

Table S9: Truncated propagation on the four held-out knockdowns of the original design (seeds 0–9). [a]Algebraic inverse; equals the infinite sum only if $\rho(J) < 1$.

Model	D=1	2	3	4	6	8	12	$(I-J)^{-1}$ [a]	$\rho(J)$
no prior	0.50	0.27	0.09	0.00	0.03	-0.01	-0.01	0.15	3.03
hard identity mask + soft site prior	0.41	0.21	0.14	0.08	0.06	0.00	-0.01	0.18	2.53
no prior, stabilised (ρ=0.8)	0.50	0.45	0.43	0.45	0.44	0.44	0.44	0.44	0.80
hard identity mask + soft site prior, stabilised (ρ=0.8)	0.41	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.80

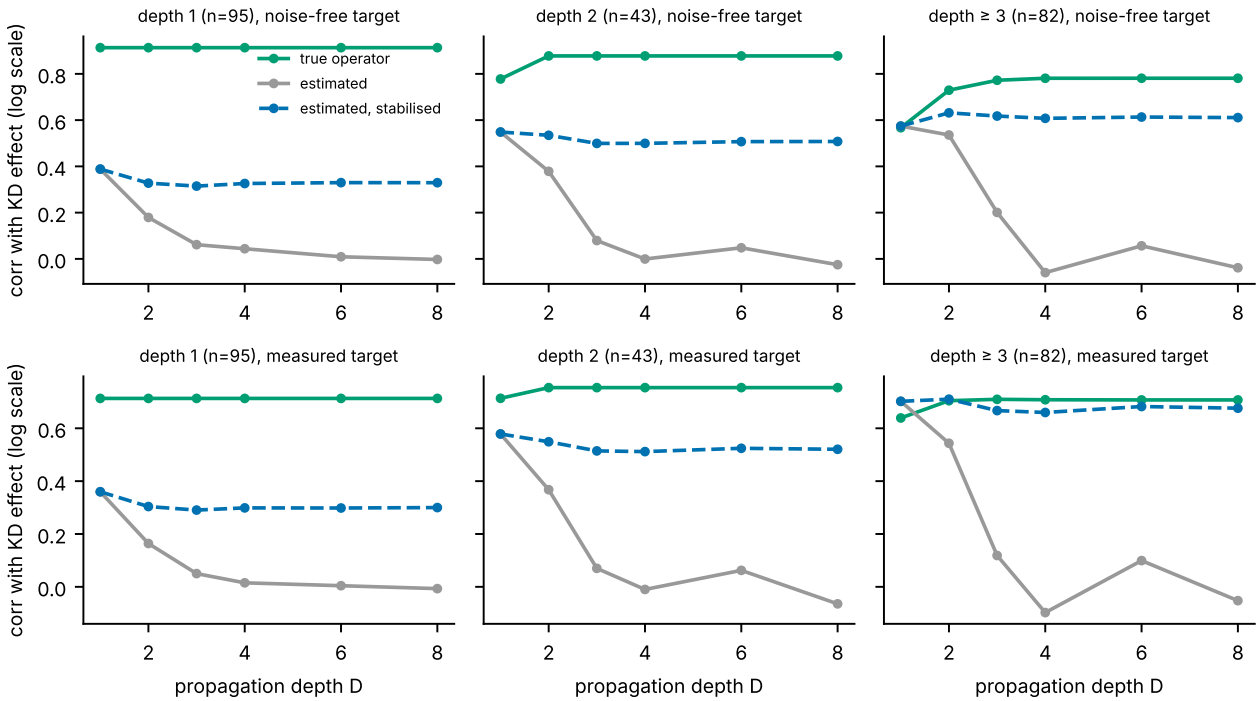


Figure S4: Propagation depth with the true vs estimated operator, both scored against the same noise-free (top) and measured (bottom) log-scale targets, stratified by each regulator's true downstream depth.

Table S10: Sign flips among strong shared targets (conditional analysis). Gain slope: total-least-squares slope of context-2 on context-1 effects.

Comparison	n	sign flips / strong shared	Jaccard	corr top-50	gain slope (median)	fold gain change
Norman 2019 (K562, CRISPRa; from earlier analysis)						
A alone vs A in C's context	260	– (1.5%)	0.185	–	–	–
A alone, split-half null	251	– (0.0%)	0.430	–	–	–
Cell-cycle state bins	152	5/10223 (0.05%)	–	0.96	–	1.26
Wessels 2023 (THP-1, Cas13 knockdown)						
A alone vs A in C's context	106	18/1657 (1.09%)	0.106	0.44	0.99	1.20
A alone, split-half null	106	1/1463 (0.07%)	0.110	0.85	1.00	1.12
Shifrut 2018 (primary T cells, CRISPR KO)						
No stimulation vs stimulation	18	0/6 (0.00%)	0.011	0.02	–	–
split-half null (no stim.)	18	0/6 (0.00%)	0.010	-0.02	–	–
split-half null (stim.)	18	0/12 (0.00%)	0.020	0.12	–	–

Table S11: Noise-constant estimators. Pooled and median per-knockdown expected-noise shares under alternative estimators of the noise constant, and a bootstrap over the 585 non-targeting controls (median estimator). Per-gene medians of squared z-scores are biased low for chi-square-like distributions and are shown divided by the χ^2_1 median (0.455).

Noise-constant estimator	pooled noise share	median per-knockdown noise share
median over controls (used)	0.470	0.737
mean over controls	0.499	0.782
20% trimmed mean	0.471	0.738
gene-specific mean	0.499	0.782
gene-specific median / chi2 median (0.455)	0.475	0.745
Bootstrap over 585 controls (95% interval, median estimator)	[0.467, 0.473]	[0.733, 0.742]

Table S12: Matched shared axes across screens ($|\cos|$ between individual axes fitted separately in each screen on the 1,457 knockdowns in both).

GW axis \ essential axis	1	2	3
1	0.95	0.17	0.08
2	0.16	0.91	0.13
3	0.05	0.12	0.02

Table S13: Positive-control robustness. Mean difference between cascade and random-mediator correlations for semi-synthetic targets with exact cascade share α (realised share equals nominal in orthogonalised generators; in the non-orthogonal generator the realised share was 0.005, 0.053, 0.102, 0.250 and 0.499). *: 95% CI lower bound above the real-data upper bound (0.004).

Generator	coverage	cascade share α (exact energy fraction)				
		0	0.05	0.1	0.25	0.5
base (orthogonal, strongest mediators, coordinates restored, equal reliability)	1.0	0.000	0.047*	0.066*	0.106*	0.153*
base (orthogonal, strongest mediators, coordinates restored, equal reliability)	0.25	-0.002	0.008*	0.013*	0.026*	0.039*
mediators ranked at random (ranking errors)	1.0	-0.003	0.006	0.009*	0.017*	0.023*
mediators ranked at random (ranking errors)	0.25	-0.003	-0.001	-0.001	0.003	0.005
mediator coordinates not restored	1.0	-0.004	-0.002	-0.002	-0.000	0.001
mediator coordinates not restored	0.25	0.000	-0.001	-0.002	-0.003	0.001
heterogeneous mediator strengths (lognormal, sd 1)	1.0	-0.003	0.035*	0.049*	0.080*	0.115*
heterogeneous mediator strengths (lognormal, sd 1)	0.25	-0.004	0.003	0.008*	0.015*	0.027*
screen-2 reliability halved	1.0	0.000	0.033*	0.046*	0.074*	0.107*
screen-2 reliability halved	0.25	-0.001	0.005	0.009*	0.018*	0.027*
screen-1 reliability halved	1.0	-0.003	0.032*	0.047*	0.074*	0.109*
screen-1 reliability halved	0.25	-0.003	0.006	0.009*	0.018*	0.026*
non-orthogonal construction (original)	1.0	-0.002	0.044*	0.064*	0.104*	0.152*
non-orthogonal construction (original)	0.25	-0.003	0.008*	0.013*	0.025*	0.038*
all adverse: random ranking, no restore, heterogeneous, screen-2 halved	1.0	-0.001	-0.003	-0.003	-0.001	-0.000
all adverse: random ranking, no restore, heterogeneous, screen-2 halved	0.25	-0.003	-0.004	-0.004	-0.001	-0.001
Real data (specific residual)	–	-0.004 [-0.012, 0.004]				

Table S14: Calibrated magnitude scores on independent-screen targets (R_0^2 uncalibrated; after training-only scalar calibration with bootstrap 95% CI; mean calibration factor $\bar{\lambda}$ over folds).

	Indep., raw			Indep., mean-centred			Indep., specific		
Predictor	R_0^2	R_0^2 calibrated [95% CI]	$\bar{\lambda}$	R_0^2	calibrated	$\bar{\lambda}$	R_0^2	calibrated	$\bar{\lambda}$
mean training profile	0.007	0.033 [0.029, 0.036]	0.53	0.000	0.000 [0.000, 0.000]	0.00	-0.000	-0.000 [-0.000, 0.000]	8.00
kNN k=10	-0.189	0.037 [0.032, 0.042]	0.28	-0.174	0.006 [0.004, 0.009]	0.15	-0.120	0.002 [0.001, 0.003]	0.10
kNN k=20	-0.117	0.042 [0.036, 0.048]	0.35	-0.106	0.007 [0.003, 0.010]	0.20	-0.056	0.003 [0.002, 0.003]	0.16
kNN k=50	-0.064	0.046 [0.040, 0.052]	0.40	-0.056	0.006 [0.002, 0.010]	0.26	-0.017	0.004 [0.003, 0.005]	0.27
kNN k=100	-0.039	0.047 [0.041, 0.052]	0.42	-0.033	0.005 [0.002, 0.008]	0.28	-0.004	0.004 [0.003, 0.005]	0.36
factorised (generic mean + pathway kNN50 + specific kNN50)	-0.017	0.043 [0.037, 0.049]	0.46	-0.016	0.012 [0.009, 0.016]	0.40	-0.017	0.004 [0.003, 0.005]	0.27

Table S15: Paired predictor comparisons (per-knockdown differences; bootstrap 95% CI).

Paired comparison	target	Δ corr [95% CI]	ΔR_0^2 calibrated [95% CI]
kNN k=100 minus kNN k=10	raw	0.041 [0.034, 0.048]	0.0097 [0.0064, 0.0127]
kNN k=100 minus kNN k=10	mean-centred	0.020 [0.009, 0.030]	-0.0011 [-0.0034, 0.0010]
kNN k=100 minus kNN k=10	specific residual	0.023 [0.016, 0.030]	0.0024 [0.0016, 0.0031]
kNN k=50 minus kNN k=10	raw	0.040 [0.033, 0.046]	0.0089 [0.0056, 0.0118]
kNN k=50 minus kNN k=10	mean-centred	0.022 [0.013, 0.031]	-0.0001 [-0.0024, 0.0020]
kNN k=50 minus kNN k=10	specific residual	0.016 [0.010, 0.022]	0.0018 [0.0011, 0.0024]
kNN k=100 minus kNN k=50	raw	0.002 [-0.001, 0.004]	0.0008 [-0.0005, 0.0022]
kNN k=100 minus kNN k=50	mean-centred	-0.002 [-0.007, 0.003]	-0.0010 [-0.0019, 0.0000]
kNN k=100 minus kNN k=50	specific residual	0.007 [0.002, 0.011]	0.0006 [0.0002, 0.0010]
factorised (generic mean + pathway kNN50 + specific kNN50) minus kNN k=50	raw	-0.007 [-0.009, -0.005]	-0.0031 [-0.0048, -0.0012]
factorised (generic mean + pathway kNN50 + specific kNN50) minus kNN k=50	mean-centred	0.005 [-0.002, 0.012]	0.0064 [0.0046, 0.0083]
factorised (generic mean + pathway kNN50 + specific kNN50) minus kNN k=50	specific residual	0.000 [-0.000, 0.000]	-0.0000 [-0.0000, 0.0000]